

ON THE ROLE OF EXTRINSIC VALUE EXCHANGE IN EXPECTATION PROPAGATION FOR CODED MIMO SYSTEMS

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ABSTRACT

This paper investigates the statistical properties of log-likelihood ratios (LLRs) at the output of an iterative signal detector based on expectation propagation (EP), highlighting the role of moment matching (MM)-based extrinsic value exchange in coded systems. Prior studies have mainly examined convergence behavior in asymptotic regimes; however, EP's advantage is not always evident in practical finite-size uncoded multi-input multi-output (MIMO) detection, where probabilistic data association (PDA), which lacks extrinsic value exchange, can perform comparably to or even better than EP. In contrast, when channel coding is applied, EP consistently outperforms PDA. To explain this, we analyze the statistical properties of demodulator output LLRs and show that extrinsic value exchange not only ensures inter-iteration independence but also enhances the Gaussianity of LLRs. We quantify this effect using negentropy and demonstrate its benefit in improving the decoder's error-correcting capability, thereby establishing the superiority of EP in coded systems.

Index Terms— Expectation propagation, LLRs, negentropy, channel decoder, coded MIMO systems

1. INTRODUCTION

Expectation propagation (EP) [1, 2, 3] is an approximate inference algorithm that achieves asymptotic Bayes-optimal performance under right-unitarily invariant observations. Its core principle is to approximate the true posterior distribution with a tractable distribution (*e.g.*, Gaussian) and iteratively update its moments to minimize the Kullback-Leibler divergence (KLD) from the true posterior. This *moment matching* (MM) operation effectively cancels self-feedback components and thereby decouples inter-iteration correlation [2, 3].

It is well known that EP without the extrinsic value generation mechanism reduces to probabilistic data association (PDA) [4, 5, 6, 7]. In underdetermined systems such as compressed sensing, where the number of unknowns exceeds the observation dimension, EP outperforms PDA [2, 3]. In contrast, under overdetermined conditions typical of conventional multi-input multi-output (MIMO) detection—where the observation dimension is much larger—both algorithms perform similarly, and PDA can even surpass EP. Thus, in uncoded MIMO detection with low spatial load, the advantage of extrinsic value exchange may appear limited.

However, in practical coded MIMO systems, the extrinsic value generation mechanism of EP becomes crucial. This is because it significantly enhances the Gaussianity of the final detector output. Soft-decision decoders are fundamentally

designed for additive white Gaussian noise (AWGN) channels and assume Gaussian-distributed input log-likelihood ratios (LLRs) [8, 9, 10]. When the input distribution deviates from the Gaussian distribution, model mismatch degrades the decoder's error-correcting performance [11]. Therefore, in iterative detection, the statistical properties of the demodulator output LLRs, determined by the final detector output, directly govern post-decoding error rates; however, this perspective has been insufficiently explored in prior works¹.

Motivated by this gap, this paper investigates how extrinsic value exchange shapes the statistical properties of demodulator output LLRs and clarifies its role in coded systems. By comparing EP and PDA, we analyze detector output statistics in both the symbol and LLR domains and quantify the impact of extrinsic value generation using negentropy, while accounting for bit-to-symbol nonlinearity. The key contribution of this work lies in a shift in focus from the *internal* convergence behavior studied in prior works [15, 16] to the output LLR statistics critical in coded systems, thereby revealing the unique role of extrinsic value exchange in EP detectors.

Notation: Set of complex numbers is denoted by $\mathbb{C}$. The real and imaginary parts of a complex quantity are denoted by $\Re[\cdot]$ and $\Im[\cdot]$, respectively. For any countable finite set $\mathcal{A}$, let $|\mathcal{A}|$ denote the number of elements in $\mathcal{A}$. Vectors are represented by lower-case boldface letters, and matrices by upper-case boldface letters. The transpose and conjugate transpose operators are denoted by $\cdot^T$ and $\cdot^H$, respectively. The $a \times a$ identity matrix is represented by $\mathbf{I}_a$. A diagonal matrix with elements of the vector $\mathbf{a}$ on its main diagonal is denoted by $\text{diag}[\mathbf{a}]$. The probability density function (PDF) of the random variable a is denoted by $p_a(\cdot)$. A complex Gaussian distribution with mean a and variance b is denoted by $\mathcal{CN}(a, b)$. The expected value of a random variable is denoted by $\mathbb{E}[\cdot]$.

2. SYSTEM MODEL

2.1. Signal Model

We consider an uplink of a multi-user MIMO (MU-MIMO) system, where a base station (BS) equipped with N antennas simultaneously serves M synchronized single-antenna user equipment (UE) devices. At the m -th UE, N_b data bits are encoded into N_c code bits with coding rate $r_c \triangleq N_b/N_c$. After interleaving, the coded bits are mapped onto frequency-

¹Note that this study considers a common system configuration where channel decoding is performed once after detection, rather than an iterative detection and decoding (IDD) scheme such as double EP [12]. In this setup, unlike IDD where decoding separates iterations [13, 14], the impact of the entire detection process is accumulated in the final demodulator output LLRs.

domain symbols² of length $K = N_c / \log_2 Q$, where each symbol is drawn from the set of Q -ary quadrature amplitude modulation (QAM) constellation points $\mathcal{X}$. The average transmit power of the modulated symbol is denoted by E_s .

We consider the k -th spatial multiplexed symbol vector $\mathbf{x}_k \triangleq [x_{1,k}, x_{2,k}, \dots, x_{M,k}] \in \mathcal{X}^{M \times 1}$. The corresponding frequency-domain received symbol vector is expressed as

$$\mathbf{y}_k \triangleq [y_{1,k}, y_{2,k}, \dots, y_{N,k}]^T = \mathbf{H}_k \mathbf{x}_k + \mathbf{z}_k, \quad (1)$$

where $\mathbf{H}_k \triangleq [\mathbf{h}_{1,k}, \mathbf{h}_{2,k}, \dots, \mathbf{h}_{M,k}] \in \mathbb{C}^{N \times M}$ is the MIMO channel matrix in the frequency domain, with $\mathbf{h}_{m,k} \in \mathbb{C}^{N \times 1}$ denoting its m -th column. The noise vector $\mathbf{z}_k \triangleq [z_{1,k}, z_{2,k}, \dots, z_{N,k}]^T \in \mathbb{C}^{N \times 1}$ consists of independent and identically distributed (i.i.d.) entries distributed as $\mathcal{CN}(0, N_0)$, where N_0 is the noise power density,

At the BS receiver, the transmitted symbol vector $\mathbf{x}_k$ is estimated from $\mathbf{y}_k$ and $\mathbf{H}_k$. The resulting soft information is then passed to a soft-decision decoder for error correction.

2.2. Algorithm Description of EP and PDA

The pseudocode of EP and PDA, designed following [2, 3, 17], is summarized in Alg. 1, where T denotes the total number of iterations and $(\cdot)^{(t)}$ denotes the iteration index. We define $\mathbf{x}_\zeta \triangleq [x_{1,\zeta}, x_{2,\zeta}, \dots, x_{M,\zeta}]^T$ and $\mathbf{V}_\zeta \triangleq \text{diag}[v_{1,\zeta}, v_{2,\zeta}, \dots, v_{M,\zeta}]$, with $\zeta \in \{A \rightarrow B, B \rightarrow A\}$. Alg. 1 consists of two modules: **Module A**, which performs soft interference cancellation and signal separation using a linear minimum mean squared error (MMSE) filter, and **Module B**, which computes the conditional expectation, *i.e.*, the general MMSE estimate, by modeling the Module A output (belief) as an AWGN channel observation. The essential distinction between PDA and EP lies in the presence of extrinsic value generation via MM [6, 7]. PDA is equivalent to EP without this process, so their first iterations are identical. The parameter α in lines 16 and 17 is a damping factor.

2.3. Bit-wise LLR calculation

Based on the message $q_{m,k,A \rightarrow B}^{(T)}(\chi)$ from Module A to B in the final iteration of Alg. 1, the LLRs of the coded bits composing the transmit symbol $\mathbf{x}_{m,k}$ are computed. Let $S \triangleq \log_2 Q$ denote the number of coded bits per symbol, and we represent them as $c_{m,k}^{(1)}, c_{m,k}^{(2)}, \dots, c_{m,k}^{(S)}$. The bit-wise LLR corresponding to $c_{m,k}^{(s)}$ is given by [14]

$$\lambda(c_{m,k}^{(s)}) \triangleq \ln \left[\frac{\sum_{\chi \in \mathcal{X} | c^{(s)}=0} q_{m,k,A \rightarrow B}^{(T)}(\chi)}{\sum_{\chi' \in \mathcal{X} | c^{(s)}=1} q_{m,k,A \rightarrow B}^{(T)}(\chi')} \right], \quad (2)$$

where $\mathcal{X} | c^{(s)} = c$ ($c \in \{0, 1\}$) denotes the subset of constellation points whose s -th bit equals c .

3. COMPARISON OF DETECTION ACCURACY

3.1. Simulation Setup

We first evaluate the impact of extrinsic value exchange on detection accuracy in terms of bit error rate (BER). The chan-

²For uncoded systems, the symbol stream is generated by directly modulating $N_b = N_c$ information bits, corresponding to $r_c = 1$.

Algorithm 1 EP and PDA

Input: $\mathbf{y}_k \in \mathbb{C}^{N \times 1}$, $\mathbf{H}_k \in \mathbb{C}^{N \times M}$, T (Num. of iterations)

Output: $q_{m,k,A \rightarrow B}^{(T)}(\chi), \forall m, \forall \chi \in \mathcal{X}$

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1:  $\mathbf{x}_{k,B \rightarrow A}^{(1)} = \mathbf{0} \in \mathbb{C}^{M \times 1}$ ,  $\mathbf{V}_{k,B \rightarrow A}^{(1)} = E_s \mathbf{I}_M$ 
2: for  $t = 1$  to  $T$  do
    /* ----- Module A ----- */
3:  $\mathbf{\Psi}_k^{(t)} = (N_0 \mathbf{I}_N + \mathbf{H}_k \mathbf{V}_{k,B \rightarrow A}^{(t)} \mathbf{H}_k^H)^{-1}$ 
4:  $\mathbf{\Xi}_k^{(t)} = (\text{diag}[\mathbf{h}_{k,1}^H \mathbf{\Psi}_k^{(t)} \mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,M}^H \mathbf{\Psi}_k^{(t)} \mathbf{h}_{k,M}])^{-1}$ 
5:  $\mathbf{x}_{k,A \rightarrow B}^{(t)} = \mathbf{x}_{k,B \rightarrow A}^{(t)} + \mathbf{\Xi}_k^{(t)} \mathbf{H}_k^H \mathbf{\Psi}_k^{(t)} (\mathbf{y}_k - \mathbf{H}_k \mathbf{x}_{k,B \rightarrow A}^{(t)})$ 
6:  $\mathbf{V}_{k,A \rightarrow B}^{(t)} = \mathbf{\Xi}_k^{(t)} - \mathbf{V}_{k,B \rightarrow A}^{(t)}$ 
7:  $\forall m, \forall \chi \in \mathcal{X} : q_{m,k,A \rightarrow B}^{(t)}(\chi) = \exp \left( - \frac{|\chi - x_{m,k,A \rightarrow B}^{(t)}|^2}{v_{m,k,A \rightarrow B}^{(t)}} \right)$ 
    /* ----- Module B ----- */
8:  $\forall m : x_{m,k,B}^{(t)} = \frac{\sum_{\chi \in \mathcal{X}} \chi \cdot q_{m,k,A \rightarrow B}^{(t)}(\chi)}{\sum_{\chi' \in \mathcal{X}} q_{m,k,A \rightarrow B}^{(t)}(\chi')}$ 
9:  $\forall m : v_{m,k,B}^{(t)} = \frac{\sum_{\chi \in \mathcal{X}} |\chi|^2 \cdot q_{m,k,A \rightarrow B}^{(t)}(\chi)}{\sum_{\chi' \in \mathcal{X}} q_{m,k,A \rightarrow B}^{(t)}(\chi')} - |x_{m,k,B}^{(t)}|^2$ 
10: if EP then
11:  $\forall m : \frac{1}{\bar{v}_{m,k,B \rightarrow A}^{(t)}} = \frac{1}{v_{m,k,B}^{(t)}} - \frac{1}{v_{m,k,A \rightarrow B}^{(t)}}$ 
12:  $\forall m : \bar{x}_{m,k,B \rightarrow A}^{(t)} = \bar{v}_{m,k,B \rightarrow A}^{(t)} \cdot \left( \frac{x_{m,k,B}^{(t)}}{v_{m,k,B}^{(t)}} - \frac{x_{m,k,A \rightarrow B}^{(t)}}{v_{m,k,A \rightarrow B}^{(t)}} \right)$ 
13: else
14:  $\forall m : \bar{v}_{m,k,B \rightarrow A}^{(t)} = v_{m,k,B}^{(t)}$ ,  $\bar{x}_{m,k,B \rightarrow A}^{(t)} = x_{m,k,B}^{(t)}$ 
15: end if
16:  $\mathbf{x}_{k,B \rightarrow A}^{(t+1)} = \alpha \bar{\mathbf{x}}_{k,B \rightarrow A}^{(t)} + (1 - \alpha) \mathbf{x}_{k,B \rightarrow A}^{(t)}$ 
17:  $\mathbf{V}_{k,B \rightarrow A}^{(t+1)} = \alpha \bar{\mathbf{V}}_{k,B \rightarrow A}^{(t)} + (1 - \alpha) \mathbf{V}_{k,B \rightarrow A}^{(t)}$ 
18: end for
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nel is modeled as a frequency-selective Rayleigh fading channel with an equal-power three-tap delay profile, assumed to be perfectly estimated at the receiver. The average received power from each transmitted antenna is assumed to be identical based on the slow open-loop power control. Gray-coded 16QAM and 64QAM are employed. As the channel code, we use the LDPC code with rate $r_c = 1/2$ and length $N_c = 1944$ specified in IEEE 802.11n [18]. Decoding is performed using the sum-product algorithm (SPA), terminated when either all parity checks are satisfied or the maximum number of iterations $T_{\text{SPA}} = 50$ is reached. The number of signal detection iterations is set to $T = 8$.

3.2. BER Performance

Fig. 1 shows the BER performance at $T \in \{1, 2, 4, 8\}$ as a function of signal-to-noise ratio (SNR), defined as $\text{SNR} \triangleq M E_s / N_0$, in a MU-MIMO system with $(N, M) = (32, 24)$.

In the uncoded MU-MIMO system in Figs. 1 (a) and (c), no significant difference is observed between EP and PDA; from the fourth iteration onward, PDA without the extrinsic value generation even outperforms EP. This implies that in finite-size overdetermined scenarios, accurate symbol separation via matrix inversion in Module A reduces the benefit of extrinsic value generation.

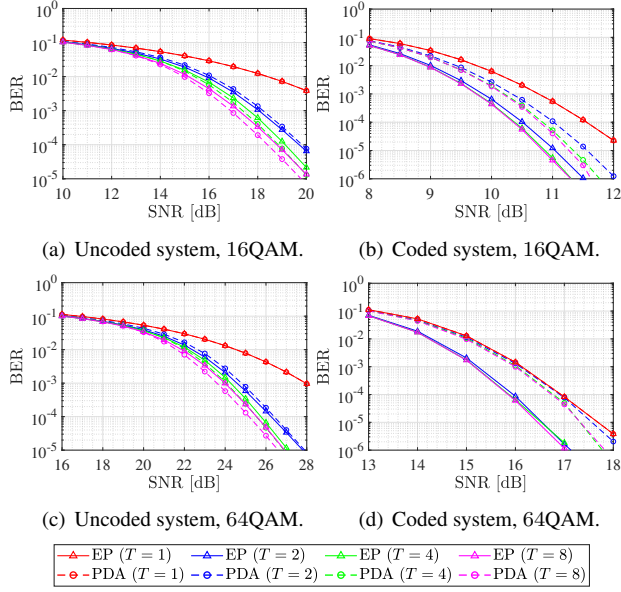


Fig. 1. BER performance versus SNR in MU-MIMO system.

In contrast, in the coded MU-MIMO system in Figs. 1 (b) and (d), EP significantly outperforms PDA, with a clear gap emerging after the second iteration, and this gain becomes particularly pronounced as the modulation order increases. Notably, the relative performance of EP and PDA is reversed between uncoded and coded systems, motivating further analysis of the statistical properties of demodulator outputs.

4. STATISTICAL PROPERTIES OF DETECTOR OUTPUT INFORMATION

We analyze a single stream within one channel realization from the previous 16QAM experiment. The detector output at the final iteration is evaluated from two perspectives: (i) the Gaussianity of the Module A output beliefs (symbol estimates), and (ii) the Gaussianity of the demodulator output LLRs provided to the soft-decision decoder. For simplicity, we analyze the real part of the symbol estimates $\Re[x_{m,k,A \rightarrow B}]$, exploiting the orthogonality of the real and imaginary components in the QAM constellation. Note that similar results hold for the imaginary part $\Im[x_{m,k,A \rightarrow B}]$.

4.1. Beliefs Output from the Detector

We first examine the distribution of equivalent noise components in the symbol domain, which offers an intuitive perspective. Fig. 2 shows the PDF of the Module A output beliefs $\Re[x_{m,k,A \rightarrow B}]$ at the final iteration ($t = 8$) with SNR = 11 dB. For Gray-coded 16QAM, the sets of symbol estimates $\mathcal{S}_i$ ($i = 1, 2, 3, 4$) are defined as

$$\mathcal{S}_i = \{\Re[x_{m,k,A \rightarrow B}], \forall m, k \mid \Re[x_{m,k}] = \chi'_i\}, \quad (3)$$

where $\chi'_i \in \{-3c, -c, c, 3c\}$ and $c = \sqrt{2E_s/3(Q-1)}$. The dotted red curves indicate the ideal Gaussian distributions with the same mean and variance as the histogram samples. Fig. 2 (a) shows that EP, through extrinsic value exchange, effectively suppresses iterative self-noise accumulation and thus preserves Gaussianity in the symbol-domain equivalent

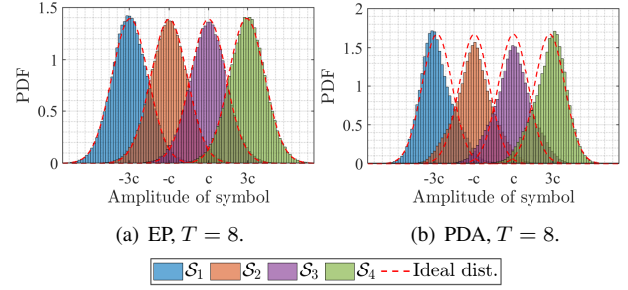


Fig. 2. PDFs of $\Re[x_{m,k,A \rightarrow B}]$ at SNR = 11 dB.

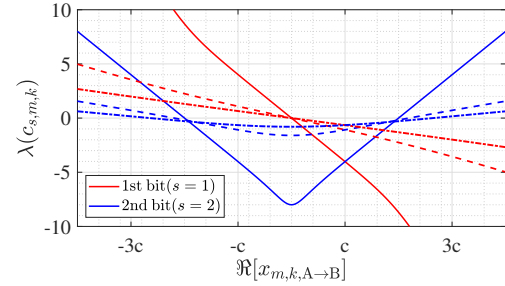


Fig. 3. Mapping function in (2) corresponding to the real part $\Re[x_{m,k,A \rightarrow B}]$ for 16QAM with $v_{m,k,A \rightarrow B} = 1, 0.5, 0.1$.

noise. In contrast, the equivalent noise distribution in PDA of Fig. 2 (b) deviates markedly from Gaussian due to self-noise accumulation, resulting in distortion.

4.2. LLRs Output from the Demodulator

Next, we evaluate how the distortion in the symbol domain translates to the bit-wise LLR domain at the demodulator output. Based on the symbol estimates in Fig. 2, the LLRs are computed by (2). For high-order QAM schemes ($Q \geq 16$), the mapping from symbol estimates to LLR varies across bit position [19], and this nonlinearity distorts the distribution of the decoder input LLRs. Fig. 3 shows the projection functions for the real part of 16QAM symbols, where the horizontal axis represents $\Re[x_{m,k,A \rightarrow B}]$ and the vertical axis represents the corresponding bit-wise LLR $\lambda(c_{s,m,k}^{(s)})$. The variance $v_{m,k,A \rightarrow B}$ in (2) is set to 1 (solid line), 0.5 (dashed line), and 0.1 (dash-dotted line). Notably, while the projection function for the first bit is monotonic, that for the second bit exhibits local minima, thereby limiting the dynamic range of the resulting LLRs. This restricts their ability to reflect true bit reliability and induces distortion in the LLR distribution [14].

Consequently, the PDF of LLRs corresponding to the first bit, $\mathcal{L}^{(1)} \triangleq \{\lambda(c_{m,k}^{(1)}), \forall m, k\}$, consists of 4 distinct components, as illustrated in Figs. 4 (a) and (c). Accordingly, $\mathcal{L}^{(1)}$ is further divided into 4 subsets as

$$\mathcal{L}_1^{(1)} \triangleq \{\lambda(c_{m,k}^{(1)}), \forall m, k \mid c_{m,k}^{(1)} = 0 \wedge \Re[x_{m,k}] = -3c\}, \quad (4a)$$

$$\mathcal{L}_2^{(1)} \triangleq \{\lambda(c_{m,k}^{(1)}), \forall m, k \mid c_{m,k}^{(1)} = 0 \wedge \Re[x_{m,k}] = -c\}, \quad (4b)$$

$$\mathcal{L}_3^{(1)} \triangleq \{\lambda(c_{m,k}^{(1)}), \forall m, k \mid c_{m,k}^{(1)} = 1 \wedge \Re[x_{m,k}] = c\}, \quad (4c)$$

$$\mathcal{L}_4^{(1)} \triangleq \{\lambda(c_{m,k}^{(1)}), \forall m, k \mid c_{m,k}^{(1)} = 1 \wedge \Re[x_{m,k}] = 3c\}, \quad (4d)$$

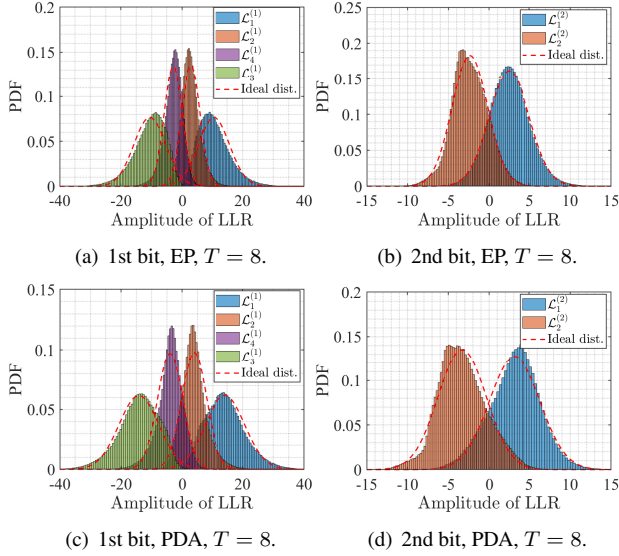


Fig. 4. PDFs of bit-wise LLRs at SNR = 11 dB.

where the PDF of $\mathcal{L}_i^{(1)}$ is shown in different colors. The dotted red line indicates the Gaussian distribution with the same mean and variance as the samples in each subset.

Similarly, the PDF of LLRs corresponding to the second bit, $\mathcal{L}^{(2)} \triangleq \{\lambda(c_{m,k}^{(2)}), \forall m, k\}$, consists of 2 components, as illustrated in Figs. 4 (b) and (d). The two subsets are defined as

$$\mathcal{L}_1^{(2)} \triangleq \{\lambda(c_{m,k}^{(2)}), \forall m, k \mid c_{m,k}^{(2)} = 0\}, \quad (5a)$$

$$\mathcal{L}_2^{(2)} \triangleq \{\lambda(c_{m,k}^{(2)}), \forall m, k \mid c_{m,k}^{(2)} = 1\}. \quad (5b)$$

In Fig. 4, distributional distortion due to the nonlinear mapping functions is observed for both EP and PDA. However, the effect is more pronounced in PDA: the first-bit LLRs exhibit excessively high kurtosis, while the second-bit LLRs show severe asymmetry. In contrast, EP maintains a distribution much closer to the expected Gaussian shape, indicating that extrinsic value exchange via MM significantly improves the statistical properties of the demodulator output LLRs.

To quantify this effect, we evaluate Gaussianity using *negentropy*, a widely used measure in independent component analysis (ICA) [20]. The negentropy of a random variable x is defined as

$$J_{\text{neg}}(x) = H_d(x_{\text{Gauss}}) - H_d, \quad (6)$$

where x_{Gauss} is a Gaussian random variable with the same mean and variance as x , and $H_d(\cdot)$ denotes the differential entropy,

$$H_d = - \int p_x(\xi) \ln p_x(\xi) d\xi. \quad (7)$$

By the maximum entropy principle, negentropy is always non-negative and equals zero only when x is Gaussian. Although accurate, computing (7) directly requires integration over the PDF, which is computationally demanding. Hence, ICA commonly uses the approximation [20]:

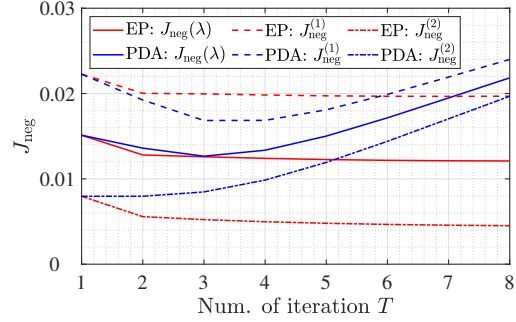


Fig. 5. Negentropy comparison between EP and PDA.

$$J_{\text{neg}}(x) \approx$$

$$k_1 \left(\mathbb{E} \left\{ \bar{x} \exp \left[-\frac{\bar{x}^2}{2} \right] \right\} \right)^2 + k_2 \left(\mathbb{E} \left\{ \exp \left[-\frac{\bar{x}^2}{2} \right] \right\} - \frac{1}{\sqrt{2}} \right)^2, \quad (8)$$

where $k_1 \triangleq 36/(8\sqrt{3} - 9)$ and $k_2 \triangleq 24/(16\sqrt{3} - 27)$, and $\bar{x} \triangleq (x - \mu_x)/\sigma_x$ with $\mu_x \triangleq \mathbb{E}[x]$ and $\sigma_x^2 \triangleq \mathbb{E}[(x - \mu_x)^2]$.

We apply (8) to the LLRs in Fig. 4. For each subset defined in (4) and (5), we first compute the negentropy values $J_{\text{neg},i}^{(1)}, i \in \{1, 2, 3, 4\}$, and $J_{\text{neg},i}^{(2)}, i \in \{1, 2\}$. Their weighted averages, taken with respect to the number of samples, yielded the first- and second-bit LLR negentropy $J_{\text{neg},i}^{(i)}, i \in \{1, 2\}$, as well as the overall LLR negentropy $J_{\text{neg}}(\lambda) \triangleq (J_{\text{neg}}^{(1)} + J_{\text{neg}}^{(2)})/2$.

Fig. 5 illustrates the evolution of the negentropy of the demodulator output LLR as a function of the number of iterations T . For EP (red curves), the negentropy decreases monotonically, indicating that Gaussianity of the LLRs is effectively preserved and even improved throughout the iterative process. In contrast, for PDA (blue curves), the negentropy initially decreases due to interference suppression, but subsequently increases sharply with iterations, clearly caused by self-feedback accumulation. These results demonstrate that the extrinsic value generation mechanism not only governs the statistical properties of beliefs exchanged within the iterations but also critically shapes the statistical behavior of the demodulator output LLRs. This underscores the previously underexplored role of extrinsic value exchange, particularly in coded wireless communication systems.

5. CONCLUSION

This paper highlighted an additional role of extrinsic value generation via MM in EP-based multi-user detection (MUD) for coded MIMO systems—beyond its well-known role in suppressing self-noise and decoupling inter-iteration dependence. Specifically, the MM mechanism governs the statistical properties of the final demodulator-output LLRs. By improving their Gaussianity, quantified by negentropy, extrinsic value exchange via MM directly enhances the effectiveness of subsequent soft-decision decoding. Consequently, although EP and PDA perform comparably in uncoded systems, EP's superior performance in coded systems is firmly rooted in this mechanism. These results underscore the need to evaluate iterative detectors not only by uncoded detection accuracy but also by the statistical quality of the information delivered to the decoder. Future work includes joint design of detector and demodulator to further improve LLR Gaussianity.

6. ACKNOWLEDGMENT

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7. REFERENCES

- [1] Thomas P. Minka, “Expectation propagation for approximate Bayesian inference,” 2013, [Online]. Available: <http://arxiv.org/abs/1301.2294>.
- [2] Sundeep Rangan, Philip Schniter, and Alyson K. Fletcher, “Vector approximate message passing,” *IEEE Trans. Inf. Theory*, vol. 65, no. 10, pp. 6664–6684, 2019.
- [3] Keigo Takeuchi, “Rigorous dynamics of expectation-propagation-based signal recovery from unitarily invariant measurements,” *IEEE Trans. Inf. Theory*, vol. 66, no. 1, pp. 368–386, 2020.
- [4] Diem Pham, Krishna Pattipati, P. Willett, and Jiankun Luo, “A generalized probabilistic data association detector for multiple antenna systems,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2004, vol. 6, pp. 3519–3522.
- [5] Shaoshi Yang, Tiejun Lv, Robert G. Maunder, and Lajos Hanzo, “From nominal to true a posteriori probabilities: An exact Bayesian theorem based probabilistic data association approach for iterative MIMO detection and decoding,” *IEEE Trans. Commun.*, vol. 61, no. 7, pp. 2782–2793, July 2013.
- [6] Takumi Takahashi, Antti Tölli, Shinsuke Ibi, and Seiichi Sampei, “Low-complexity large MIMO detection via layered belief propagation in beam domain,” *IEEE Trans. Wireless Commun.*, vol. 21, no. 1, pp. 234–249, 2022.
- [7] Takuma Kobayashi, Takumi Takahashi, Shinsuke Ibi, and Seiichi Sampei, “Concatenated structure of EP and turbo equalizer for overloaded massive MIMO detection,” in *Proc. 21st IEEE Annu. Consum. Commun. Netw. Conf. (CCNC)*, 2024, pp. 690–695.
- [8] Sae-Young Chung, T.J. Richardson, and R.L. Urbanke, “Analysis of sum-product decoding of low-density parity-check codes using a Gaussian approximation,” *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 657–670, 2001.
- [9] Lajos L. Hanzo, T. H. Liew, B. L. Yeap, R. Y. S. Tee, and Soon Xin Ng, *Turbo Coding, Turbo Equalisation and Space-Time Coding: EXIT-Chart-Aided Near-Capacity Designs for Wireless Channels*, Wiley-IEEE Press, 2011.
- [10] Mohammad Rowshan, Min Qiu, Yixuan Xie, Xinyi Gu, and Jinhong Yuan, “Channel coding toward 6G: Technical overview and outlook,” *IEEE Open J. Commun. Soc.*, vol. 5, pp. 2585–2685, 2024.
- [11] Mirsad Cirkic, Daniel Persson, Jan-Åke Larsson, and Erik G. Larsson, “Approximating the LLR distribution for a class of soft-output MIMO detectors,” *IEEE Trans. Signal Process.*, vol. 60, no. 12, pp. 6421–6434, 2012.
- [12] Juan José Murillo-Fuentes, Irene Santos, José Carlos Aradillas, and Matilde Sánchez-Fernández, “A low-complexity double EP-based detector for iterative detection and decoding in MIMO,” *IEEE Trans. Commun.*, vol. 69, no. 3, pp. 1538–1547, 2021.
- [13] Mohamad A. Ahmed and Charalampos C. Tsimenidis, “Tight upper bound performance of full-duplex MIMO-BICM-IDD systems in the presence of residual self-interference,” *IEEE Trans. Wireless Commun.*, vol. 17, no. 1, pp. 520–532, 2018.
- [14] Fuga Kobayashi, Takumi Takahashi, Shinsuke Ibi, and Hideki Ochiai, “EXIT chart analysis of expectation propagation-based iterative detection and decoding,” *IEEE Open J. Commun. Soc.*, vol. 6, pp. 5249–5268, 2025.
- [15] Junjie Ma and Li Ping, “Orthogonal AMP,” *IEEE Access*, vol. 5, pp. 2020–2033, 2017.
- [16] Tomoharu Furudoi, Takumi Takahashi, Shinsuke Ibi, and Hideki Ochiai, “Discrete-valued signal estimation via low-complexity message passing algorithm for highly correlated measurements,” 2024, [Online]. Available: <https://arxiv.org/abs/2411.07558>.
- [17] J. Céspedes, P. M. Olmos, M. Sánchez-Fernández, and F. Perez-Cruz, “Expectation propagation detection for high-order high-dimensional MIMO systems,” *IEEE Trans. Commun.*, vol. 62, no. 8, pp. 2840–2849, Aug. 2014.
- [18] “IEEE standard for information technology—telecommunications and information exchange between systems local and metropolitan area networks—specific requirements - part 11: Wireless LAN medium access control (MAC) and physical layer (PHY) specifications,” *IEEE Std 802.11-2016 (Revision of IEEE Std 802.11-2012)*, pp. 1–3534, 2016.
- [19] Daichi Shirase, Takumi Takahashi, Shinsuke Ibi, Kazushi Muraoka, Naoto Ishii, and Seiichi Sampei, “Negentropy-aware loss function for trainable belief propagation in coded MIMO detection,” in *Proc. IEEE GLOBECOM*, 2021, pp. 1–6.
- [20] Aapo Hyvärinen, Jarmo Hurri, and Patrik O. Hoyer, *Independent Component Analysis*, Adaptive and Cognitive Dynamic Systems: Signal Processing, Learning, Communications and Control. Wiley, 2004.